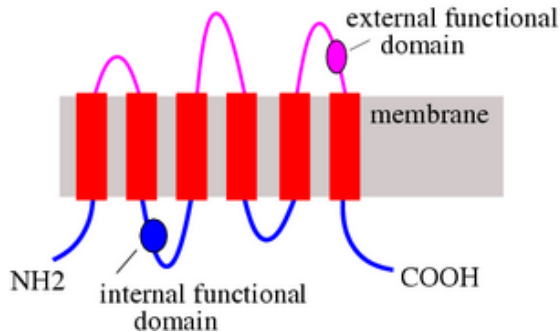


Data Wrangling in Bioinformatics Assignment 3

Hidden Markov Models. Mass spectrometry pre-processing and identification

1. Receptor proteins typically make good drug targets. These proteins weave in and out of the membrane as follows: intracellular, transmembrane, extracellular, transmembrane, and so on (see figure below).



The transmembrane domain contains mainly hydrophobic residues (L,I,V,F,W) and each is equally likely. All other residues occur with low probability $\frac{1}{100}$. The extra-cellular domain contains predominantly polar residues R,K,D,E,Q,N all equally likely, with the remaining residues appearing with probability $\frac{1}{100}$. Finally, all residues are equally likely in the intra-cellular domain. Describe an HMM $M = (Q, \Sigma, \pi, e, T)$ for detecting transmembrane proteins. The description should include a graphical and text explanation of Q, Σ, π, e, T . Assume that transition from any state to itself is always 0.6. All other permitted transitions are equally likely.

2. In class, we worked on the problem of *de novo* identification. Here, we will consider the other problem of identifying peptides against a database. Also, we will consider the case of ‘top-down’ mass spectrometry, which is done on a pure protein sample, or on a sample of low complexity (small number of proteins). In this protocol, no extraneous peptidase is used for digestion. Instead, endogenously digested peptides are fragmented and the MS2 spectrum acquired with very high accuracy. Thus, in top-down MS, we have high accuracy fragments that are not preferentially *b, y*, but can also be *a, c, x, z*; the parent peptide may be much longer than a typical tryptic peptide, and the parent peptide and fragment ions may have higher charge values. The mgf file provided to you (mgf is a text readable format) is the top-down MS2 spectrum of a peptide with charge 7. The fragment ions may have charge from 1 to 6.
 - a. (5 points) Write a script to calculate the theoretical mass of a peptide sequence. What is the theoretical residue mass of ‘PEPTIDE’? What is the theoretical mass of ‘PEPTIDE’? What is the precursor (parent) ion mass if the charge was $Z = 2$?
 - b. (10 points) Calculate the amino-acid mass and the residue mass of the parent peptide given the m/z value and charge state in the mgf file (show work). Why is high mass accuracy important?
 - c. (5 points) Generate candidate peptides from the fasta file CAH_BOVIN that are within a mass tolerance of 30ppm from the precursor ion mass. What is the equation used to calculate mass accuracy?
 - d. (10 points) For each candidate peptide, generate its theoretical spectrum given that each fragment ion can have a charge from +1 to +6 and be of type *a, b, c, x, y*, or *z*. Output a table containing charge 2 ions *a, b, y*. **Hint.** You can check the correctness of your answers on test peptides at the Protein Prospector website (MS-Product)
 - e. (15 points) Come up with a spectral similarity score between two spectra (i.e., theoretical vs. experimental) - taking into account both m/z and intensity values. Choose an appropriate fragment ion mass tolerance. Write pseudocode and clearly state the input and output of your approach.

- f. (15 points)** Which peptide from the fasta file CAH_BOVIN best explains the input spectrum? Plot the experimental spectrum, highlighting the ions from your candidate peptide that best explain the most intense peaks. Also indicate the ion type, charge, and sequence for the top three most intense peaks.